

Problem Sheet

Geometric and Topological Data Domains

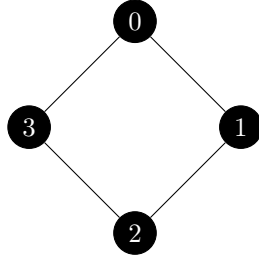
Topology and Geometry in Machine Learning

Lecture 2

Remark. The exercises are marked in increasing difficulty by *, **, ***.

Exercise 1 (Graphs, the Laplacian, and the Graph Fourier transform)

Let $G = (V, E)$ be the 4-cycle with $V = \{0, 1, 2, 3\}$ and $E = \{\{0, 1\}, \{1, 2\}, \{2, 3\}, \{3, 0\}\}$.



- (a, *) Write down the adjacency matrix A , the degree matrix D , and the graph Laplacian $L = D - A$ of G .
- (b, *) Choose an orientation of the four edges and write the signed incidence matrix $B \in \mathbb{R}^{4 \times 4}$. Verify that $L = BB^T$ and that $B^T \mathbf{1} = \mathbf{0}$ (recall that $\mathbf{1} = (1, 1, 1, 1)^T$).
- (c, **) Compute the eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \lambda_4$ of the Laplacian. Find the Fiedler value λ_2 and interpret it. Compute a basis of orthogonal eigenvectors of the Laplacian $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.
- (d, ***) For a general graph $G = (V, E)$ and a signal $\mathbf{f} \in \mathbb{R}^n$, prove the identity

$$\mathbf{f}^T L \mathbf{f} = \sum_{\{i,j\} \in E} (f_i - f_j)^2.$$

Deduce that L is positive semi-definite and that $\mathbf{f}^T L \mathbf{f} = 0$ if and only if $\mathbf{f}$ is constant on each connected component.

- (e, *) Let $\mathbf{f} = (4, 0, 0, 0)^T$ be a spike signal at vertex 0. Compute the Graph Fourier coefficients $a_i = \mathbf{u}_i^T \mathbf{f}$ and check that $\mathbf{f} = \sum_{i=1}^4 a_i \mathbf{u}_i$.
- (f, **) Apply the low-pass filter that keeps only the modes with $\lambda_i \leq 2$. Write the filtered signal $\mathbf{f}_{\text{lp}}$ in vertex coordinates and compare with $\mathbf{f}$. What do you get if you keep only the mode $\lambda_1 = 0$?

Solution.

- (a) Ordering vertices 0, 1, 2, 3,

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \quad D = 2I_4, \quad L = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

- (b) Orient $e_1 = (0, 1), e_2 = (1, 2), e_3 = (2, 3), e_4 = (3, 0)$; with $B_{ve} = -1$ at the head and $+1$ at the tail,

$$B = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 \end{pmatrix}.$$

- (c) The eigenvalues of the Laplacian are $\lambda_1 = 0, \lambda_2 = \lambda_3 = 2, \lambda_4 = 4$. They are obtained as the roots of:

$$\det(L - \lambda I) = \det \begin{pmatrix} 2 - \lambda & -1 & 0 & -1 \\ -1 & 2 - \lambda & -1 & 0 \\ 0 & -1 & 2 - \lambda & -1 \\ -1 & 0 & -1 & 2 - \lambda \end{pmatrix} = \lambda(\lambda - 2)^2(\lambda - 4) = 0.$$

The Fiedler value is $\lambda_2 = 2$: it is bounded away from 0, reflecting that C_4 is well connected (a single edge cut disconnects it into two paths, which is why it is not larger).

- (d) With any orientation, $\mathbf{f}^\top L \mathbf{f} = \mathbf{f}^\top B B^\top \mathbf{f} = \|B^\top \mathbf{f}\|^2 = \sum_{e=\{i,j\} \in E} (f_i - f_j)^2 \geq 0$, using that the e -entry of $B^\top \mathbf{f}$ is $f_{\text{head}(e)} - f_{\text{tail}(e)}$. Thus $L \succeq 0$. Moreover $\mathbf{f}^\top L \mathbf{f} = 0$ iff $f_i = f_j$ for every edge, i.e. iff $\mathbf{f}$ is constant along every path, i.e. iff $\mathbf{f}$ is constant on each connected component.

- (e) Solving $(L - \lambda I)\mathbf{u} = 0$ for each λ :

- $\lambda = 0$: $L\mathbf{u} = 0$ forces $u_1 = u_2 = u_3 = u_4$, giving $\mathbf{u}_1 = \frac{1}{2}(1, 1, 1, 1)$.
- $\lambda = 4$: $(L - 4I)\mathbf{u} = 0$ gives $u_1 = u_3$ and $u_2 = u_4 = -u_1$, giving $\mathbf{u}_4 = \frac{1}{2}(1, -1, 1, -1)$.
- $\lambda = 2$: $(L - 2I)\mathbf{u} = 0$ reduces to $u_1 + u_3 = 0$ and $u_2 + u_4 = 0$, so the eigenspace is $\{(s, t, -s, -t) : s, t \in \mathbb{R}\}$, of dimension 2. Taking the orthogonal pair $(s, t) = (1, 1)$ and $(1, -1)$,

$$\mathbf{u}_2 = \frac{1}{2}(1, 1, -1, -1), \quad \mathbf{u}_3 = \frac{1}{2}(1, -1, -1, 1).$$

The $\mathbf{u}_i$ are orthonormal and each has first entry $\frac{1}{2}$.

- (f) $a_i = \mathbf{u}_i^\top \mathbf{f} = \frac{1}{2} \cdot 4 = 2$ for every i , since each $\mathbf{u}_i$ has first entry $\frac{1}{2}$. Indeed $2(\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3 + \mathbf{u}_4) = (4, 0, 0, 0)^\top = \mathbf{f}$.

- (g) Keeping $\lambda_i \leq 2$ drops the $\mathbf{u}_4$ term:

$$\mathbf{f}_{\text{lp}} = 2\mathbf{u}_1 + 2\mathbf{u}_2 + 2\mathbf{u}_3 = (1, 1, 1, 1) + (1, 1, -1, -1) + (1, -1, -1, 1) = (3, 1, -1, 1)^\top.$$

The energy that sat entirely on vertex 0 has been spread to its neighbours 1 and 3; the signal is smoother (smaller edge differences) and the spike is attenuated from 4 to 3. Keeping only $\lambda_1 = 0$ gives $\mathbf{f}_{\text{DC}} = 2\mathbf{u}_1 = (1, 1, 1, 1)$, the mean value of $\mathbf{f}$ broadcast to every vertex.

Exercise 2 (Simplicial complexes and homology)

For each simplicial complex X below, orient every simplex by increasing vertex label.

Linear algebra toolbox. The following linear algebra facts can be used throughout the exercise:

- for a quotient vector space V/W , $\dim V/W = \dim V - \dim W$;
- the rank of matrix is the number of linearly independent rows (or columns), in other words, the dimension of its image;
- $\text{rank } \partial_k + \dim \ker \partial_k = \dim C_k = |X_k|$ (from rank-nullity).

(a, *) **Hollow vs. filled triangle.** Let T° have vertices $\{0, 1, 2\}$ and edges $\{(0, 1), (1, 2), (0, 2)\}$ (no 2-simplex), and let T be T° together with the 2-simplex $(0, 1, 2)$.

- (i) Write the boundary matrices ∂_1 (for both) and ∂_2 (for T) and verify $\partial_1\partial_2 = 0$.
- (ii) Compute H_0, H_1, H_2 and the Betti numbers $\beta_0, \beta_1, \beta_2$ of T° and of T .

(b, *) **Wedge of two circles.** Let W have

$$V = \{0, 1, 2, 3, 4\}, \quad E = \{(0, 1), (1, 2), (0, 2), (2, 3), (3, 4), (2, 4)\}$$

(two triangle boundaries glued at the vertex 2, nothing filled). Compute all Betti numbers of W .

(c, **) **Hollow tetrahedron.** Let $S = \partial\Delta^3$ be the boundary of the 3-simplex on $\{0, 1, 2, 3\}$: all 4 vertices, all 6 edges, all 4 triangles, no 3-simplex. Compute $\dim C_k(S)$, the ranks of ∂_1, ∂_2 , and all Betti numbers. Identify a generator of $H_2(S)$ explicitly as a chain.

Solution.

- (a) (i) A basis of C_2 is $\{(0, 1, 2)\}$ (for the filled triangle, for the hollow one this space is trivial), a basis of C_1 is $\{(0, 1), (1, 2), (0, 2)\}$ and a basis of C_0 is $\{\{0\}, \{1\}, \{2\}\}$. With respect to these, the boundary maps have matrices

$$\partial_1 = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix}, \quad \partial_2(0, 1, 2) = (1, 2) - (0, 2) + (0, 1) \implies \partial_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

and indeed

$$\partial_1\partial_2 = (1 + 0 - 1, -1 + 1 + 0, 0 - 1 + 1)^\top = 0.$$

- (ii) For T° : $\text{rank } \partial_1 = 2$, so $\dim \ker \partial_1 = 1$ and, with $\partial_2 = 0$, $H_1 = \ker \partial_1 \cong \mathbb{R}$, $\beta_1 = 1$ (generated by the cycle $(0, 1) + (1, 2) - (0, 2)$); $H_0 = \mathbb{R}^3 / \text{im } \partial_1 \cong \mathbb{R}$, $\beta_0 = 1$; $H_2 = 0$.

For T : $\text{im } \partial_2$ is 1-dimensional and contained in the 1-dimensional $\ker \partial_1$, so $H_1 = \ker \partial_1 / \text{im } \partial_2 = 0$, $\beta_1 = 0$; ∂_2 is injective so $H_2 = \ker \partial_2 = 0$; $\beta_0 = 1$ unchanged. Filling the triangle makes the single 1-cycle a boundary ($\text{im } \partial_2$), so it becomes trivial in H_1 .

- (b) $|W_0| = 5$, $|W_1| = 6$, $|W_2| = 0$. W is connected so $\beta_0 = 1$ and $\text{rank } \partial_1 = 5 - 1 = 4$; hence $\dim \ker \partial_1 = 6 - 4 = 2$ and, since $\partial_2 = 0$, $\beta_1 = \dim H_1 = 2$ (one class per loop). $\beta_k = 0$ for $k \geq 2$.
- (c) $\dim C_0 = 4$, $\dim C_1 = 6$, $\dim C_2 = 4$, $C_3 = 0$. S is connected: $\text{rank } \partial_1 = 3$, $\beta_0 = 1$, $\dim \ker \partial_1 = 3$. The 4 oriented faces satisfy exactly one linear relation, $\partial_2((1, 2, 3) - (0, 2, 3) + (0, 1, 3) - (0, 1, 2)) = 0$, so $\dim \ker \partial_2 = 1$ and $\text{rank } \partial_2 = 3$. Then $\dim \text{im } \partial_2 = 3 = \dim \ker \partial_1$ and $\text{im } \partial_2 \subseteq \ker \partial_1$, so $H_1 = 0$, $\beta_1 = 0$. Finally $H_2 = \ker \partial_2 \cong \mathbb{R}$ (as $\partial_3 = 0$), $\beta_2 = 1$, with generator the fundamental class $z = (1, 2, 3) - (0, 2, 3) + (0, 1, 3) - (0, 1, 2)$. $\beta_k = 0$ for $k \geq 3$.

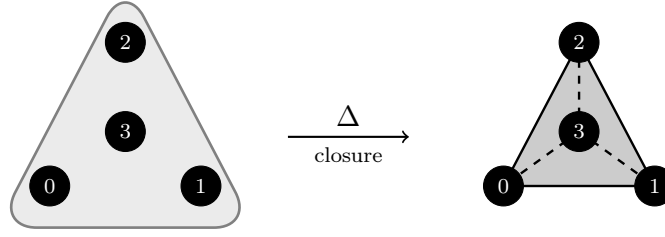
Exercise 3 (Simplicial complexes, hypergraphs, and closures)

Recall that a hypergraph is a pair $H = (V, \mathcal{E})$ with elements in $\mathcal{E}$ being non-empty subsets of V ; and that a (finite abstract) simplicial complex is a hypergraph that is *closed under inclusion*: $\sigma \in \mathcal{E}$ and $\emptyset \neq \tau \subseteq \sigma$ imply $\tau \in \mathcal{E}$. Define the *downward closure* (or *simplicial closure*) of H by

$$\Delta(H) := \{ \tau : \emptyset \neq \tau \subseteq h \text{ for some } h \in \mathcal{E} \},$$

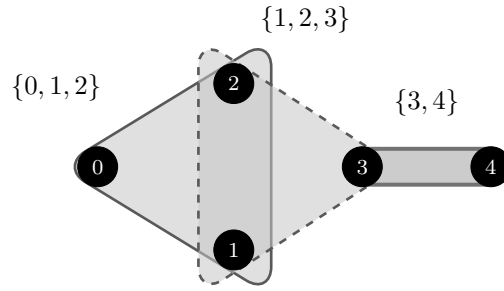
or in other words, by adding all non-empty subsets of the hyperedges of H .

For example, for a hypergraph on 4 vertices with only one hyperedge $h = \{0, 1, 2, 3\}$, the closure is the full tetrahedron Δ^3 :



$$H = (\{0, 1, 2, 3\}, \{\{0, 1, 2, 3\}\}) \quad \Delta(H) = \Delta^3 \text{ (all } 2^4 - 1 = 15 \text{ faces)}$$

- (a, **) Show that $\Delta(H)$ is a simplicial complex, and that $\Delta(H) = H$ if and only if H is already a simplicial complex. Show also $\Delta(\Delta(H)) = \Delta(H)$.
- (b *) Let $V = \{0, 1, 2, 3, 4\}$ and $\mathcal{E} = \{\{0, 1, 2\}, \{1, 2, 3\}, \{3, 4\}\}$. List $\Delta(H)$ by dimension (X_0, X_1, X_2) and give its dimension.



- (c, ***) **Universal property.** Prove that for every simplicial complex X on V ,

$$\Delta(H) \subseteq X \iff \mathcal{E} \subseteq X.$$

Conclude that $\Delta(H)$ is the smallest simplicial complex containing H .

- (d, **) If a hyperedge h has $|h| = k$ vertices, how many simplices does it contribute to $\Delta(H)$? What would be more efficient, memory wise: to keep all simplices or to only keep the hyperedges?

Solution.

- (a) Pick $\tau \in \Delta(H)$ and a subset $\emptyset \neq \rho \subseteq \tau \in \Delta(H)$. By definition of $\Delta(H)$, there exists $h \in \mathcal{E}$ with $\tau \subseteq h$; then $\rho \subseteq h$, so $\rho \in \Delta(H)$. Hence $\Delta(H)$ is closed under inclusion, i.e. a simplicial complex.

We now prove that $\Delta(H) = H \iff H$ is a simplicial complex.

($\implies$) If $\Delta(H) = H$ then H equals a simplicial complex, so it is one.

($\impliedby$) Let $H = (V, \mathcal{E})$ be a closed under inclusion. For $h \in \mathcal{E}$, by definition, $h \in \Delta(H)$, so in general, $\mathcal{E} \subseteq \Delta(H)$. Now, for $\tau \in \Delta(H)$ a proper subset of vertices, we know that $\tau \subseteq h$ for some $h \in \mathcal{E}$. Since H is closed under inclusion, $\tau \in \mathcal{E}$. Hence $\Delta(H) = H$.

Since $\Delta(H)$ is already a simplicial complex, applying the property that we have just prove, $\Delta(\Delta(H)) = \Delta(H)$.

- (b) The complex has simplices

$$X_0 = \{0, 1, 2, 3, 4\}, \quad X_1 = \{\{0, 1\}, \{0, 2\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{3, 4\}\},$$

$$X_2 = \{\{0, 1, 2\}, \{1, 2, 3\}\}$$

and dimension $\dim \Delta(H) = 2$.

(c)

($\implies$) $\mathcal{E} \subseteq \Delta(H) \subseteq X$.

($\impliedby$) Pick $\tau \in \Delta(H)$. By definition, there exists $h \in \mathcal{E} \subset X$ with $\tau \subseteq h$. Since X is closed under inclusion, $\tau \in X$, and $\Delta(H) \subseteq X$.

This property means that any simplicial complex X containing H contains $\Delta(H)$. Since $\Delta(H)$ is already a simplicial complex containing H , it is the minimum complex in that family.

(d) For a hyperedge h with $|h| = k$, the number of non-empty subsets is $2^k - 1$, so h contributes $2^k - 1$ simplices to $\Delta(H)$. Storing all faces is exponential in the hyperedge size; so it is more efficient to store the top-dimensional interaction without its entire power set.

Exercise 4 (Proving the Hodge decomposition theorem)

In the lecture, we have established that the boundary matrices of a complex satisfy $\partial_k \partial_{k+1} = 0$. This identity is the only thing one needs to prove the Hodge decomposition theorem.

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ be real matrices such that

$$AB = 0.$$

These can be seen as maps $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $B : \mathbb{R}^p \rightarrow \mathbb{R}^n$ between Euclidean spaces, where every space carries the standard inner product (so the adjoint of a matrix is its transpose), and we set

$$\Delta = A^\top A + BB^\top \in \mathbb{R}^{n \times n}.$$

Remark. If $A = \partial_k$ and $B = \partial_{k+1}$ are the boundary matrices of a simplicial complex, then $\Delta = L_k$ is the k -th Hodge Laplacian.

Linear algebra toolbox. You can use the following facts about matrices throughout the exercise. For $M \in \mathbb{R}^{r \times s}$ and $x \in \mathbb{R}^s$, $y \in \mathbb{R}^r$:

- $\langle Mx, y \rangle = (Mx)^\top y = x^\top M^\top y = \langle x, M^\top y \rangle$;
- *rank-nullity*: $\dim \ker M + \text{rank } M = s$ (number of columns), and $\text{rank } M = \text{rank } M^\top$;
- for any subspace $W \subseteq \mathbb{R}^s$, $\mathbb{R}^s = W \oplus W^\perp$ orthogonally;
- if $W \subseteq U$ are subspaces of $\mathbb{R}^s$, then $U = W \oplus (U \cap W^\perp)$ orthogonally.

Part 1: relating the image and kernel of $M \in \mathbb{R}^{r \times s}$ and its transpose $M^\top \in \mathbb{R}^{s \times r}$.

- (1, *) Prove $\ker M^\top = (\text{im } M)^\perp$ and $\ker M = (\text{im } M^\top)^\perp$.
- (2, *) Deduce the orthogonal splittings $\mathbb{R}^r = \text{im } M \oplus \ker M^\top$ and $\mathbb{R}^s = \text{im } M^\top \oplus \ker M$.
- (3, **) Prove $\ker(M^\top M) = \ker M$. (*Hint: if $M^\top Mx = 0$, look at $x^\top M^\top Mx$.*)
- (4, **) Using rank-nullity, prove $\text{rank}(M^\top M) = \text{rank } M^\top$ and hence $\text{im}(M^\top M) = \text{im } M^\top$.

Part 2: properties of two matrices with $AB = 0$.

- (5, **) Show $AB = 0 \iff \text{im } B \subseteq \ker A$. Taking transposes, show also $\text{im } A^\top \subseteq \ker B^\top$, and conclude $\text{im } B \perp \text{im } A^\top$.
- (6, **) For $x \in \mathbb{R}^n$ prove $x^\top \Delta x = \|Ax\|^2 + \|B^\top x\|^2$, and deduce

$$\ker \Delta = \ker A \cap \ker B^\top.$$

- (7, **) Prove the orthogonal splitting $\ker A = \text{im } B \oplus \ker \Delta$.

Hint: $\text{im } B \subseteq \ker A$ by (5); apply the last toolbox fact with $W = \text{im } B$, $U = \ker A$, then rewrite $(\text{im } B)^\perp$ and $\ker A \cap \ker B^\top$.

(8, *) Prove likewise $\ker B^\top = \operatorname{im} A^\top \oplus \ker \Delta$.

Hint: apply (7) verbatim to the pair $B^\top, A^\top$ in place of A, B ; note this does not change Δ .

(9, **) **Hodge decomposition.** Combine (2) (applied to B) with (8) to obtain the orthogonal direct sum

$$\boxed{\mathbb{R}^n = \operatorname{im} A^\top \oplus \ker \Delta \oplus \operatorname{im} B}$$

and, using that Δ is symmetric, $\operatorname{im} \Delta = \operatorname{im} A^\top \oplus \operatorname{im} B$.

Solution. Over $\mathbb{R}$ with the dot product, $\langle u, v \rangle = u^\top v$ and the adjoint of a matrix is its transpose.

(1) $y \in \ker M^\top \iff M^\top y = 0 \iff \langle M^\top y, x \rangle = 0, \forall x \in \mathbb{R}^s \iff \langle y, Mx \rangle = 0, \forall x \in \mathbb{R}^s \iff y \perp \operatorname{im} M$. So $\ker M^\top = (\operatorname{im} M)^\perp$.

Replacing M by $M^\top$ gives $\ker M = (\operatorname{im} M^\top)^\perp$.

(2) From the toolbox, since $\operatorname{im} M \subset \mathbb{R}^r$, we have $\mathbb{R}^r = \operatorname{im} M \oplus (\operatorname{im} M)^\perp$ and by (1), $\mathbb{R}^r = \operatorname{im} M \oplus \ker M^\top$. Similarly, for $M^\top$, $\mathbb{R}^s = \operatorname{im} M^\top \oplus \ker M$.

(3) ($\subseteq$) If $x \in \ker M$, then $Mx = 0$, so $M^\top Mx = M^\top 0 = 0$, hence $x \in \ker(M^\top M)$.

($\supseteq$) If $M^\top Mx = 0$ then $0 = x^\top M^\top Mx = \|Mx\|^2$, so $Mx = 0$; hence $\ker(M^\top M) \subseteq \ker M$.

(4) $M^\top M \in \mathbb{R}^{s \times s}$ has s columns, so by rank-nullity, (3) and $\operatorname{rank} M = \operatorname{rank} M^\top$ we get,

$$\operatorname{rank}(M^\top M) = s - \dim \ker(M^\top M) = s - \dim \ker M = \operatorname{rank} M = \operatorname{rank} M^\top.$$

Since $\operatorname{im}(M^\top M) \subseteq \operatorname{im} M^\top$ and the two have equal dimension, they are equal.

(5)

($\implies$) For $y \in \operatorname{im} B$, $\exists x \in \mathbb{R}^p$ such that $y = Bx$. Then $Ay = ABx = 0$, so $y \in \ker A$. Hence $\operatorname{im} B \subseteq \ker A$.

($\impliedby$) If $\operatorname{im} B \subseteq \ker A$, then for any $x \in \mathbb{R}^p$, $ABx = A(Bx) = 0$. Hence $AB = 0$.

Taking transposes, $(AB)^\top = B^\top A^\top = 0$, so by the same statement $\operatorname{im} A^\top \subseteq \ker B^\top$.

By (1), $\ker B^\top = (\operatorname{im} B)^\perp$, hence $\operatorname{im} A^\top \subseteq (\operatorname{im} B)^\perp$, and thus $\operatorname{im} A^\top \perp \operatorname{im} B$.

(6) For $x \in \mathbb{R}^n$, we have

$$x^\top \Delta x = x^\top A^\top A x + x^\top B B^\top x = \|Ax\|^2 + \|B^\top x\|^2 \geq 0.$$

We now prove that $\ker \Delta = \ker A \cap \ker B^\top$.

($\subseteq$) If $x \in \ker \Delta$, then $0 = x^\top \Delta x = \|Ax\|^2 + \|B^\top x\|^2$, so $Ax = 0$ and $B^\top x = 0$, hence $x \in \ker A \cap \ker B^\top$.

($\supseteq$) If $x \in \ker A \cap \ker B^\top$, then $Ax = 0$ and $B^\top x = 0$, so $x^\top \Delta x = \|Ax\|^2 + \|B^\top x\|^2 = 0$, hence $x \in \ker \Delta$.

(7) By (5), $\operatorname{im} B \subseteq \ker A$.

The toolbox fact gives $\ker A = \operatorname{im} B \oplus (\ker A \cap (\operatorname{im} B)^\perp)$, orthogonally.

Now $(\operatorname{im} B)^\perp = \ker B^\top$ by (1), and $\ker A \cap \ker B^\top = \ker \Delta$ by (6). So $\ker A = \operatorname{im} B \oplus \ker \Delta$.

(8) Put $A' = B^\top$, $B' = A^\top$.

Then $A'B' = B^\top A^\top = 0$ by (5), and $\Delta' = A'^\top A' + B'B'^\top = BB^\top + A^\top A = \Delta$.

Applying (7) to A' and B' : $\ker A' = \operatorname{im} B' \oplus \ker \Delta'$, i.e. $\ker B^\top = \operatorname{im} A^\top \oplus \ker \Delta$.

(9) By (2) applied to $B \in \mathbb{R}^{n \times p}$:

$$\mathbb{R}^n = \text{im } B \oplus \ker B^\top.$$

Substituting (8),

$$\mathbb{R}^n = \text{im } B \oplus (\text{im } A^\top \oplus \ker \Delta) = \text{im } A^\top \oplus \ker \Delta \oplus \text{im } B.$$

The three summands are mutually orthogonal.

Finally Δ is symmetric, so $\text{im } \Delta = (\ker \Delta)^\perp = \text{im } A^\top \oplus \text{im } B$.